BSIDESNYC 2026 JOHN JAY COLLEGE · NEW YORK

Cleartext at the *Edge*

A Neptune R900 case study in RF telemetry exposure

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LEAD PRODUCT SECURITY ENGINEER

RX 915.000 MHZ · RECEIVE ONLY



WHOAMI

Rick Ramirez

NOW	Lead product security engineer: OT/ICS, medical devices, cyber-physical systems
BEFORE	U.S. Marine Corps radio and electronics technician
RESEARCH	Passive RF telemetry, smart utility systems, cyber-physical privacy
CERTIFIED	GICSP: Global Industrial Cyber Security Professional

01 · SETUP

Scope and Boundaries

THIS TALK IS

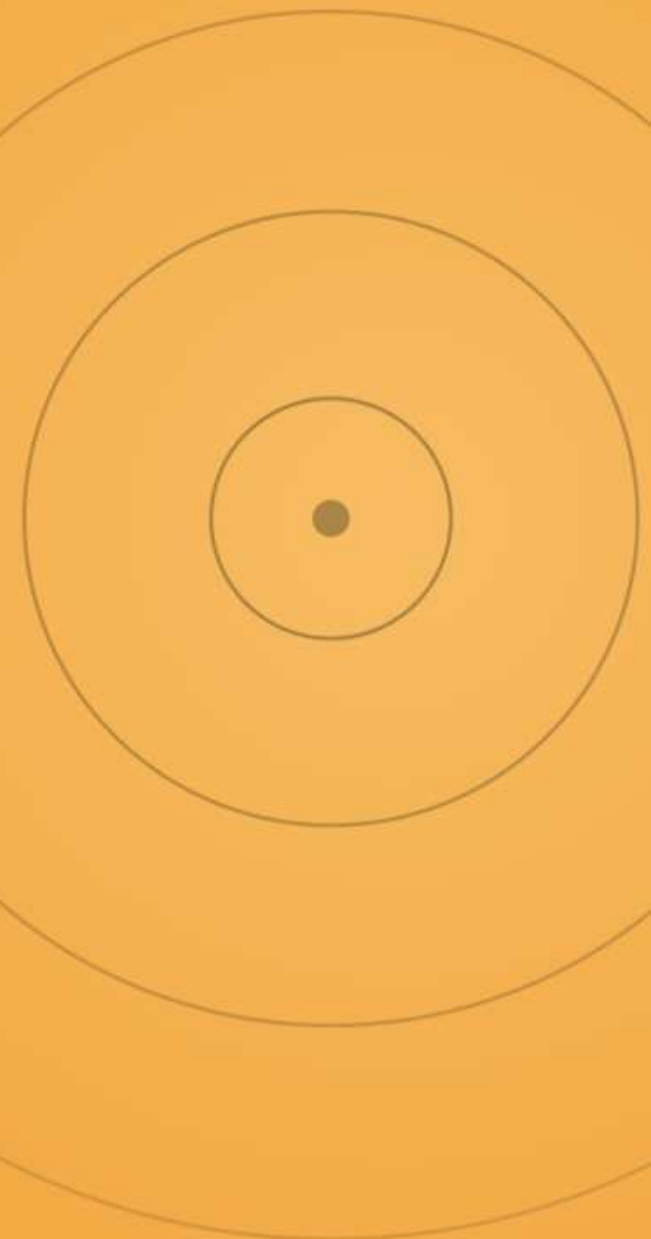
- ✓ A passive RF case study
- ✓ A privacy and security analysis
- ✓ A look at systemic design risk
- ✓ A case for RF as attack surface

THIS TALK IS NOT

- – A guide to targeting meters
- – A release of anyone’s telemetry
- – An attempt to disrupt utilities
- – An exploitation walkthrough

THE PREMISE

The boundary is not the firewall. *The boundary is wherever the signal can be heard.*



02

CHAPTER 02 · RF

Primer

Why radio sits outside security visibility, and how a low-cost receiver makes it readable.

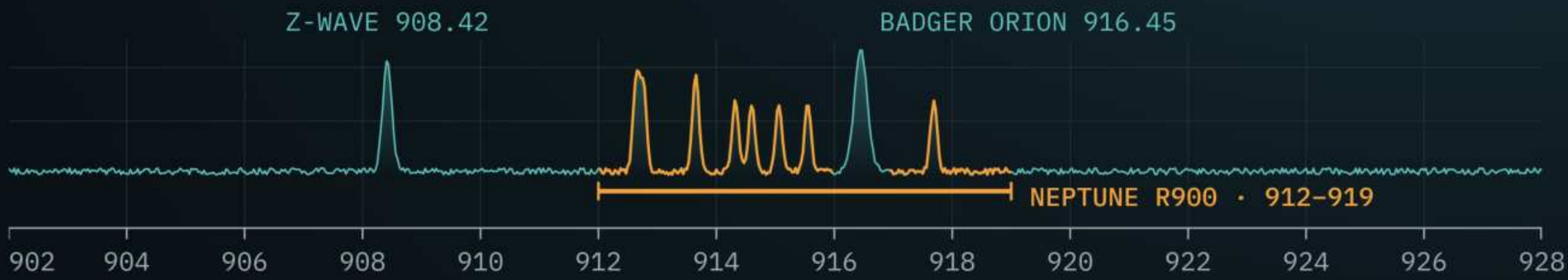
The Blind Spot

Security programs are built around the packet. RF happens before there is a packet.

If a system transmits, someone else may be able to receive it. None of your tools will record that they did.

CONTROL	NETWORK	RF BROADCAST
Firewall logs	Sees it	Never sees it
EDR	Sees it	Never sees it
SIEM	Sees it	Never sees it
Packet capture	Sees it	Never sees it
Asset inventory / NAC	Sees it	Never sees it

The 902–928 MHz Band



WHO SHARES IT

- Water meters
- Gas & electric meters
- SCADA radios
- Home automation
- Weather & soil sensors
- LoRaWAN nodes

$30–60

An RTL-SDR dongle, a wire antenna and open-source rtl_433 hear all of it.

03

CHAPTER 03 · NEPTUNE R900

Case Study

How cleartext water-meter telemetry surfaced in a garden sensor’s console.



CLEARTEXT AT THE EDGE

CH 08 · 910.67 MHZ

How This Started

01

I was building a soil-moisture sensor for my garden, on 915 MHz.

02

While cleaning up its output, other packets appeared in the console.

03

They were water meters. Mine and my neighbors'. In plaintext.

Not new: cleartext AMR was documented in 2012 (Rouf et al., ACM CCS). In 2026, it is still on the air.

No targeting. No special tools. It simply showed up.



NEPTUNE MACH 10 · R900i
SERIAL MASKED

03 · CASE STUDY

Neptune R900

ROLE	Radio endpoint that reports a water meter’s reading
BAND	902–928 MHz ISM, frequency hopping
CADENCE	Mobile message about every 14 s; fixed-network message every 7.5 min
IDENTITY	Static ID tied to the meter’s serial number
PROTECTION	No encryption. No message authentication.
LIFESPAN	10–20 years in service

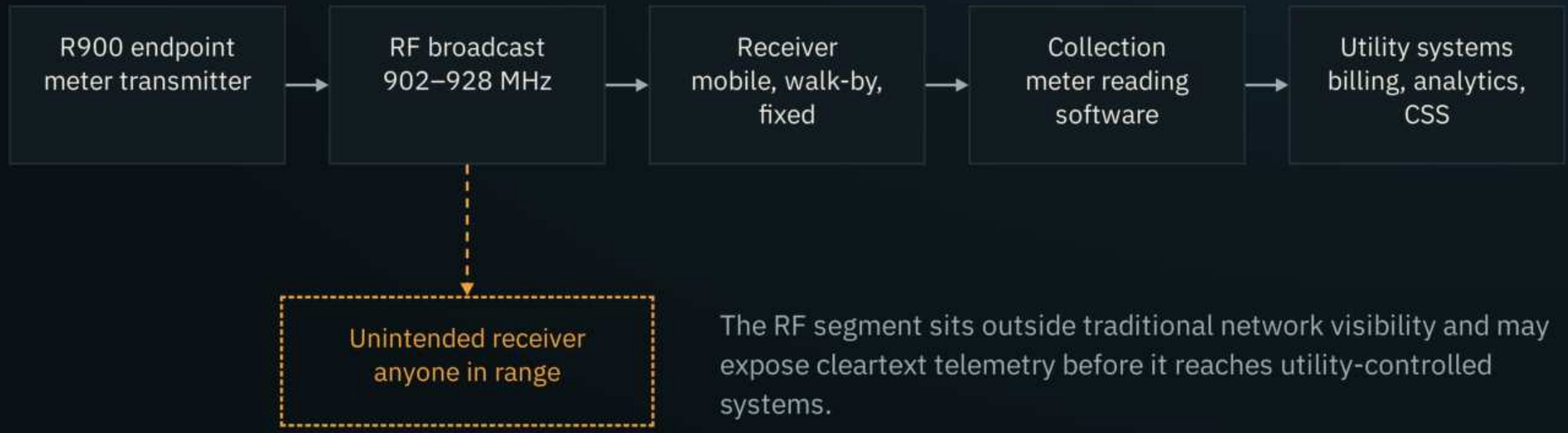
Sources: Neptune product literature; City of Oneida, NY meter memo, Feb 2026

03 · CASE STUDY

AMR Data Flow

RF SEGMENT · NO NETWORK VISIBILITY

UTILITY-CONTROLLED SYSTEMS



The RF segment sits outside traditional network visibility and may expose cleartext telemetry before it reaches utility-controlled systems.

What a Decoded Frame Can Reveal

EXAMPLE JSON OUTPUT · FIELD NAMES ONLY

```
{
  "model": "Neptune-R900",
  "id": "Unique Meter ID",
  "consumption": "Water Usage Amount",
  "leak": 0,
  "backflow": 0,
  "rssi": "Signal Strength",
  "time": "Time signal was transmitted"
}
```

IDENTITY

A meter ID, static and tied to the serial number

USAGE

A consumption reading, repeated over time

STATUS

Leak and backflow or tamper-style flags, where present

SIGNAL

Signal strength and a timestamp

03 · CASE STUDY

Demo: Passive RF to Structured Telemetry

STEP 1 OF 5 A water meter transmits a short radio burst.

SPECTRUM · 902–928 MHZ



WHAT THE RECEIVER GETS · SIMULATED, SANITIZED

Burst detected at 912.5 MHz

Duration: a few milliseconds

■ water meter ■ other devices

1 SIGNAL	2 DEMODULATE	3 BITS	4 DECODE	5 FIELDS
hear a burst	wave → symbols	symbols → 1s, 0s	find the frame	readable values

CLEARTEXT AT THE EDGE

CH 13 · 916.86 MHZ

03 · CASE STUDY

Demo: Passive RF to Structured Telemetry

STEP 2 OF 5 The receiver turns the radio wave into symbols.

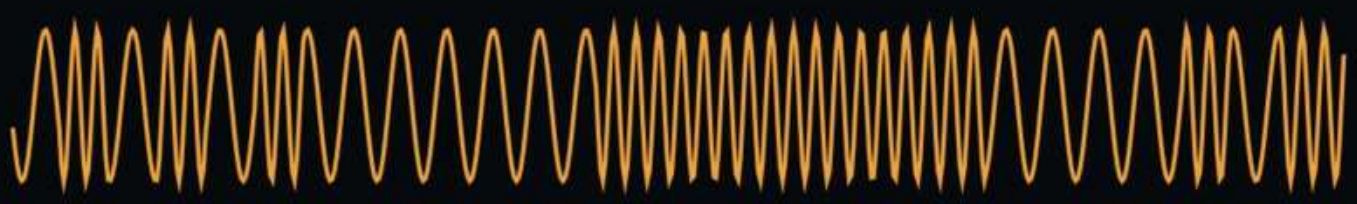
SPECTRUM · 902–928 MHZ



■ water meter ■ other devices

WHAT THE RECEIVER GETS · SIMULATED, SANITIZED

Frequency shifts = symbols



1 SIGNAL	2 DEMODULATE	3 BITS	4 DECODE	5 FIELDS
hear a burst	wave → symbols	symbols → 1s, 0s	find the frame	readable values

CLEARTEXT AT THE EDGE

CH 13 · 916.86 MHZ

03 · CASE STUDY

Demo: Passive RF to Structured Telemetry

STEP 3 OF 5 Symbols become a raw stream of 1s and 0s.

SPECTRUM · 902–928 MHZ



WHAT THE RECEIVER GETS · SIMULATED, SANITIZED

```
0 1 0 1 0 1 0 0 0 0 0 1 1 1 1 1 1 1 1 0 0 0 1 0 1 1 0 1 1 0
1 0 0 1 0 1 1 1 0 0 1 0 1 0 1 0 1 1 0 0 1 0 0 1 1 0 1 1 1 1
1 1 1 0 0 0 1 1 1 0 1 1 1 1 0 0 0 0 0 1 0 0 1 1 1 0 0 1 1 1
```

■ water meter ■ other devices

1 SIGNAL	2 DEMODULATE	3 BITS	4 DECODE	5 FIELDS
hear a burst	wave → symbols	symbols → 1s, 0s	find the frame	readable values

CLEARTEXT AT THE EDGE

CH 13 · 916.86 MHZ

03 · CASE STUDY

Demo: Passive RF to Structured Telemetry

STEP 4 OF 5 The decoder finds the frame and checks it is intact.

SPECTRUM · 902–928 MHZ



WHAT THE RECEIVER GETS · SIMULATED, SANITIZED

```
0 1 0 1 0 1 0 0 0 0 0 0 1 1 1 1 1 1 1 1 0 0 0 1 0 1 1 0 1 1 0
1 0 0 1 0 1 1 1 0 0 1 0 1 0 1 0 1 1 0 0 0 1 0 0 1 1 0 1 1 1 1
1 1 1 0 0 0 1 1 1 0 1 1 1 1 0 0 0 0 0 1 0 0 1 1 1 0 0 1 1 1 1
```

frame start found checksum OK ✓

■ water meter ■ other devices

1 SIGNAL	2 DEMODULATE	3 BITS	4 DECODE	5 FIELDS
hear a burst	wave → symbols	symbols → 1s, 0s	find the frame	readable values

CLEARTEXT AT THE EDGE

CH 13 · 916.86 MHZ

03 · CASE STUDY

Demo: Passive RF to Structured Telemetry

STEP 5 OF 5 Named fields appear. No key or password needed.

SPECTRUM · 902–928 MHZ



■ water meter ■ other devices

WHAT THE RECEIVER GETS · SIMULATED, SANITIZED

Model	Neptune-R900	
Meter ID	••••4821	never changes → trackable
Water reading	0043529	rises over time → routine
Leak flag	0	
Backflow flag	0	

1 SIGNAL
hear a burst

2 DEMODULATE
wave → symbols

3 BITS
symbols → 1s, 0s

4 DECODE
find the frame

5 FIELDS
readable values

CLEARTEXT AT THE EDGE

CH 13 · 916.86 MHZ

03 · CASE STUDY

Everyone in Range Hears It



CLEARTEXT AT THE EDGE

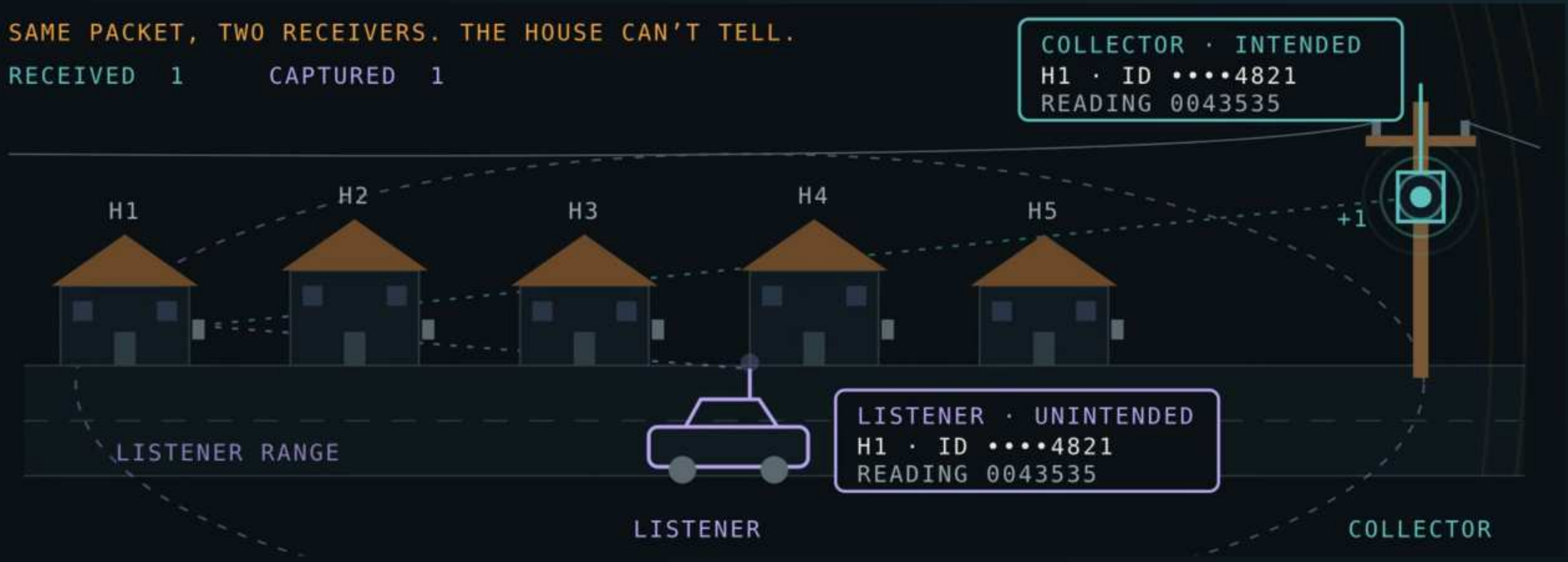
CH 14 · 918.10 MHZ

03 · CASE STUDY

Everyone in Range Hears It

SAME PACKET, TWO RECEIVERS. THE HOUSE CAN'T TELL.

RECEIVED 1 CAPTURED 1

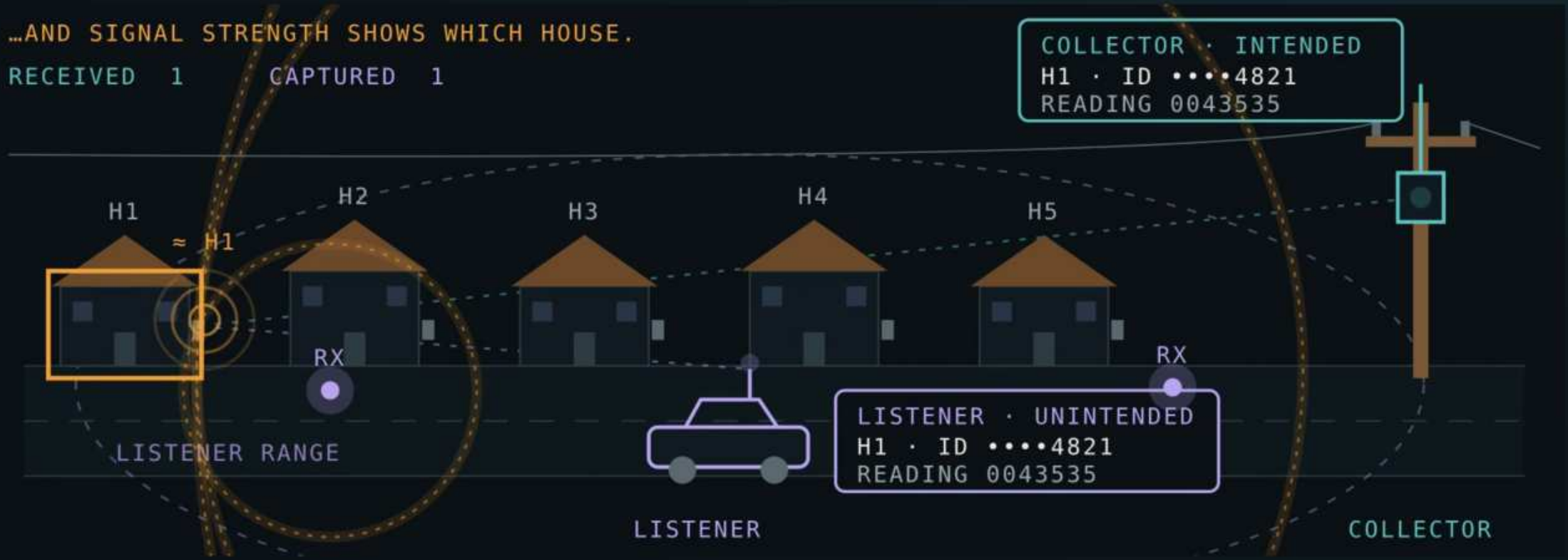


CLEARTEXT AT THE EDGE

CH 14 · 918.10 MHZ

03 · CASE STUDY

Everyone in Range Hears It



04

CHAPTER 04 · PATTERN OF LIFE

Risk

What the telemetry reveals, and how the exposure rates.

Telemetry Can Become Pattern-of-Life Data





DAILY ROUTINES

Usage can reveal occupancy patterns and daily life

INDUSTRIAL SITES

Consumption can hint at manufacturing, treatment, or other sensitive processes

VACANT AND AWAY

Usage patterns show when a building is likely empty

Potential inferences from repeated consumption data

CVSS vs. Context

V1 · CLEARTEXT · RISKS 1–6

8.3 High

CVSS:3.1/AV:A/AC:L/PR:N/UI:N
S:U/C:H/I:N/A:N · CR:H

V2 · NO AUTH · RISK 7

5.0 Medium

CVSS:3.1/AV:A/AC:L/PR:N/UI:N
S:U/C:N/I:L/A:N · IR:H

#	RISK	CVSS	CONTEXT	RATING
1	Passive surveillance	8.3	20 5×4	High
2	Static identifiers	8.3	20 5×4	High
3	Cleartext telemetry	8.3	20 5×4	High
4	Pattern of life	8.3	16 4×4	High
5	Asset enumeration	8.3	12 4×3	Medium–High
6	Weak trust boundaries	8.3	16 4×4	High
7	Integrity concerns	5.0	12 3×4	Medium–High

CVSS 3.1 environmental, utility context · Context = L × I, 1–5 each

Both methods rate it High.

05

CHAPTER 05 · BEYOND ONE METER

Implications

Why it matters, what was disclosed, and what security teams should do.

Disclosure Scope

FEB 20, 2026

Cleartext R900 packets observed while testing a garden sensor

FEB 2026

Reported to CISA

DURING PROCESS

Outreach to Neptune and other vendors: most never responded

MAY 2026

CISA closed the report without further action

BROADER DISCLOSURE · AMR/AMI VENDORS

Itron

Gas, electric

Neptune

Water, sewer. Primary case study

Badger Meter

Water, sewer

Landis+Gyr

Electric. Sent a Customer Information Letter to impacted customers

What Security Teams Should Take Away

PROGRAM & PROCUREMENT

01 Treat RF telemetry as an external-facing interface

02 Include RF in threat modeling and architecture reviews

03 Add RF security requirements to procurement language

04 Run privacy impact assessments for AMR/AMI deployments

05 Monitor for abnormal RF behavior where feasible

DESIGN CONTROLS

- ✓ Encrypt sensitive telemetry payloads
- ✓ Authenticate messages
- ✓ Add replay protection
- ✓ Avoid static identifiers where possible
- ✓ Reduce unnecessary broadcast fields

THE TAKEAWAY

RF is part of *the attack surface.*

Security reviews should include the signal layer.

QUESTIONS & DISCUSSION · END OF TRANSMISSION

Thank *you.*

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